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XO Arena — Performance Plan v2

v1 lives at `doc/archive/Performance_Plan.md` and is closed out (17 phases, Ready ~330ms floor on the small surface set that existed at the time).

This v2 is a fresh pass against the *current* platform: Fly.io hosting, four services (backend / landing / tournament / packages), SSE+POST realtime, Tables-as-primitive, Multi-Skill Bots (Phase 3.8), the Intelligent Guide journey, and a mobile UI. v1's numbers and assumptions are stale; this plan re-baselines first and prescribes fixes second.

Goal: bear down on real *and* perceived performance across **every** user- facing surface — desktop and mobile, cold and warm, anonymous and signed-in, free play and tournament play.

Targets (binding budgets)

Every page on Fly prod, p75 over a 7-day RUM window:

Metric	Desktop	Mobile (4G mid-range)
FCP	≤ 100ms	≤ 250ms
Ready (spinner gone)	≤ 200ms	≤ 500ms
LCP	≤ 200ms	≤ 500ms
INP (interaction p75)	≤ 100ms	≤ 200ms
TTFB (HTML)	≤ 60ms	≤ 150ms

Backend p95 budgets (per endpoint, single-region):

Endpoint family	p95
`GET /bots`, `/leaderboard`, `/puzzles`	≤ 30ms
`GET /tournaments`, `/tournaments/:id`	≤ 100ms
`GET /users/*` (stats, history)	≤ 80ms
`POST /rt/tables` (create)	≤ 120ms
`POST /rt/tables/:slug/move`	≤ 60ms
SSE channel join → first event	≤ 80ms

Bundle budgets per route:

- Initial parse on `/`: ≤ 120 KB gz JS, ≤ 20 KB CSS
- Largest other route: ≤ +60 KB gz over baseline
- Hero image: ≤ 200 KB on mobile (today: `colosseum-bg.jpg` is **888 KB** — see Phase 3)

If a phase's measured win doesn't move at least one budgeted metric, mark it *landed but ineffective* and revisit the assumption.

Current baseline (2026-05-02 staging, cold-anon)

Full numbers + caveats: `doc/Performance_Snapshot_2026-05-02.md`. The single-pass median across 13 routes:

Metric	Target (desktop)	Measured	Over by
FCP	≤ 100ms	~525ms	5.3×
LCP	≤ 200ms	~640ms	3.2×
Ready	≤ 200ms	~760ms	3.8×

Metric	Target (mobile)	Measured	Over by
FCP	≤ 250ms	~1360ms	5.4×
LCP	≤ 500ms	~1450ms	2.9×
Ready	≤ 500ms	~1580ms	3.2×

Every route is over budget on both devices. Three load-bearing facts from the snapshot drive the priorities below:

1. **One bundle, every page.** First-paint JS is ~493 KB gz on every route (`vendor-react` 82 KB + `main.supported` 411 KB). The `main.supported` chunk is **1,529 KB raw / 411 KB gzip** and contains the entire app — no per-route lazy splitting at the page level today. FCP / LCP / Ready collapse to within ~100ms of each other on every route, the fingerprint of a single bundle blocking the first paint.
2. **PlayVsBot is genuinely 600–700ms slower than the next-worst route** on both devices (10-run confirmation). Not noise. The `/play?action= vs-community-bot` path runs sequential: `getCommunityBot()` → `/api/v1/rt/tables` POST → redirect, all before the spinner clears.
3. **Image bytes were not counted.** `colosseum-bg.jpg` (888 KB) didn't appear in the static byte total — likely loads async after Ready resolves. Need to confirm before promoting Phase 3.

The data does not support chasing DB indexes (Phase 2), SSE latency (Phase 5), or cross-service hops (Phase 4) yet — the bundle + parse floor is currently dominating everything else by a ~10× margin. Those phases stay in the plan but move to Tier 2.

Prod sanity-check baseline (2026-05-04)

To confirm staging is a usable proxy for prod, ran `perf-v2.js` against `xo-landing-prod.fly.dev` (`--target=prod --warmup`, 5 runs × 13 routes × desktop+mobile, cold-anon), then immediately re-ran against `xo-landing-staging.fly.dev` to remove staleness from the comparison.

Saved at:

- `perf/baselines/perf-prod-2026-05-04T10-49-09-639Z.json`
- `perf/baselines/perf-staging-2026-05-04T10-55-42-186Z.json`

Headline: prod ≈ staging within run-to-run noise. Staging is a usable proxy.

Device	Mean Δ% (prod − staging, ready p50)	Notes
Desktop	**+8.6%** (n=13)	Two routes prod-faster (Stats −8%); rest +2 to +19%
Mobile	**−0.1%** (n=13)	Spar prod-faster by 15%; everything else within ±5%

Per-route ready-p50 (ms), prod vs same-day staging:

Route	Desktop prod	Desktop stg	Δ%	Mobile prod	Mobile stg	Δ%
Home	985	878	+12%	2077	2036	+2%
Play	969	849	+14%	2051	2046	+0%
PlayVsBot	1490	1421	+5%	2061	2055	+0%
Leaderboard	968	811	+19%	2076	2053	+1%
Puzzles	850	796	+7%	2127	2051	+4%
Tournaments	906	767	+18%	2069	2059	+0%
Tables	977	883	+11%	2085	2038	+2%
Spar	860	837	+3%	1714	2012	−15%
Stats	843	918	−8%	2046	2053	−0%
Profile	913	811	+13%	2110	2052	+3%
ProfileBots	901	830	+9%	2055	2037	+1%
Gym	860	846	+2%	2059	2077	−1%
Settings	893	826	+8%	2081	2063	+1%

> **Earlier draft of this section reported +23%** because the staging comparison was a 2-day-old baseline (2026-05-02) — staging machines had been warm during that earlier sweep. A same-clock comparison shows mobile is essentially identical and desktop is within first-route cold-start variance. Lesson for the playbook: **always run both environments back-to-back** when comparing.

What this means for the plan

- **Staging is a valid proxy for prod.** Optimizations that move staging numbers will move prod numbers by approximately the same amount. Phase 1/1b acceptance can be measured on staging alone going forward; re-baseline prod once per Tier (not per phase) for confirmation.
- **The bundle + parse floor still dominates on prod.** Same JS weight (490 KB), same FCP/LCP/Ready collapse pattern. Phase 1 (per-route splitting) is still the right next move.
- **PlayVsBot tax is environment-independent.** Both prod and staging show PlayVsBot ~600ms above the rest at desktop p50. Phase 1b's sequential-init flattening applies equally to both.
- **One outlier still worth investigating:** Tournaments desktop p95 hit 6448ms on staging (single run) and 3543ms on prod's earlier run — a cold backend machine handling the first `/api/tournaments` request. `--warmup` only HEADs landing. *Fix:* extend `--warmup` to also hit `/api/version` on backend and `/api/tournaments` on tournament before measuring. File against Phase 6 (cold-start).

Follow-ups still open

- Run a one-off `Home × 5` against `aiarena.callidity.com` to confirm custom-domain TLS does not add measurable latency.
- Re-baseline prod after the next Tier-0 phase lands to confirm the staging-as-proxy assumption holds as the bundle gets smaller.

Pre-flight — before running Phase 0

Cold benchmark numbers are only meaningful if the environment is real and the data is representative. Run this checklist before every measurement pass.

Environment up

All three Fly apps + the staging DB must be reachable:

```
fly status -a xo-backend-staging \
  && fly status -a xo-landing-staging \
  && fly status -a xo-tournament-staging
curl -sf https://xo-backend-staging.fly.dev/api/version && echo OK
curl -sf https://xo-tournament-staging.fly.dev/api/health && echo OK
curl -sfI https://xo-landing-staging.fly.dev/ | head -1
```

Warm the machines (don't measure cold-start as cold-page)

Fly auto-suspends idle staging machines. The first request after suspend includes a multi-second machine boot — that's a *cold-start* number, not a cold-page number, and it'll skew every percentile. Two options:

- **Recommended for a benchmark window:** disable auto-suspend on each app for the duration of the run, then re-enable.

  ```
  for app in xo-backend-staging xo-landing-staging xo-tournament-staging; do
    fly scale count 1 --max-per-region=1 -a "$app"  # ensure 1 machine running
    # disable auto_stop_machines in fly.toml or use the API; revert after
  done
  ```

- **Or:** discard the first run of each route in the script (the warmup pass is built into `perf/perf.js` Phase 0.1 work).

Test data shape

Heavy pages need representative rows. Staging may be sparse. Before the run, seed (or confirm) at least:

- One in-progress 32-participant cup (for `TournamentDetailPage` worst-case).
- One user with 5+ bots, at least 2 of which have a trained ML skill (for `/profile?section=bots` and `/gym` heavy paths).
- ≥ 50 leaderboard entries with a mix of bots + humans (for `/leaderboard` disambiguation render cost).
- ≥ 1 active table (`/tables` list) and 1 spectator-joinable table.

If the data isn't there, document it in the snapshot — *don't pretend the numbers reflect prod-shape*.

Auth

`cold-signed-in` and `warm-signed-in` runs need a working test account on staging:

```
# Required env vars (same ones e2e/qa.mjs uses):
echo "$TEST_USER_EMAIL"   # must resolve
echo "$TEST_USER_PASSWORD"

# Smoke the login flow once before the benchmark:
cd e2e && BASE_URL=https://xo-landing-staging.fly.dev npx playwright test smoke --project=chromium
```

Quiesce other traffic

If anyone else is QAing on staging during the run, the numbers are noise. Coordinate a window or run after-hours.

Snapshot the build

Capture the staging git SHA so the snapshot is reproducible:

```
curl -s https://xo-backend-staging.fly.dev/api/version  # logs commit + version
```

Record it at the top of `Performance_Snapshot_<date>.md`.

Post-flight — clean up after the run

Every change made for the benchmark must be reverted, or the next person to hit staging gets a polluted environment (cost from pinned machines, polluted metrics from seed data, locked test users, etc.). Run this checklist *every* time, immediately after the snapshot is captured.

Restore Fly machine policy

If you disabled auto-suspend for the warm pass, put it back:

```
for app in xo-backend-staging xo-landing-staging xo-tournament-staging; do
  # Either revert fly.toml + redeploy, or use the API to flip
  # auto_stop_machines back to "stop". Confirm via:
  fly status -a "$app"   # machines should return to suspended on idle
done
```

If you scaled machine count up, scale back down.

Remove seeded benchmark data

Anything created *for the benchmark* must be deleted. The `um` CLI is the canonical path so cascades fire correctly (Tournament-Participant rows get cleaned up before User delete, etc.):

```
# Test cup created for TournamentDetailPage worst-case
docker compose exec -T backend node --experimental-transform-types --no-warnings \
  src/cli/um.js tournament:delete <cup-id>

# Bots minted for the heavy /profile + /gym runs
docker compose exec -T backend node --experimental-transform-types --no-warnings \
  src/cli/um.js bot:delete <botId>   # repeat per seeded bot

# Throwaway users (anything tagged isTestUser:true that isn't the persistent
# QA account) — confirm before bulk delete
```

If a Phase 0 helper script seeded the data, it should ship with a paired `--cleanup` flag so the teardown is one command, not a manual list.

Reset RUM ingest

Phase 0.2 wires `web-vitals` → `POST /api/v1/rum`. The benchmark's synthetic runs will dump hundreds of fake samples into `RumSample` and skew the next day's percentiles. Either:

- Tag every benchmark sample with `source: 'synthetic-baseline'` and exclude it from RUM dashboards by default; or
- Delete the rows the run produced:

  ```
  DELETE FROM "RumSample"
  WHERE "createdAt" >= '<run-start-ts>'
    AND "createdAt" <= '<run-end-ts>';
  ```

Same logic for backend RED metrics if they're persisted (Prometheus scrapes are ephemeral, so usually nothing to do there).

Restore CI / config drift

If you flipped any flag or env var on staging for the run (e.g. enabled extra logging, raised a cache TTL to test something, turned off a feature gate), revert before walking away. List them in the snapshot doc so nothing's forgotten.

Sanity check

One last pass to confirm staging is back to its normal shape:

```
fly status -a xo-backend-staging              # auto_stop policy normal
curl -s https://xo-backend-staging.fly.dev/api/version | grep -i version
docker compose exec -T backend node --experimental-transform-types --no-warnings \
  src/cli/um.js list --filter=isTestUser     # only persistent QA rows remain
```

Smoke the e2e suite once to confirm nothing regressed during cleanup:

```
cd e2e && BASE_URL=https://xo-landing-staging.fly.dev npx playwright test smoke --project=chromium
```

If anything fails, the snapshot is suspect — re-run after fixing.

Phase 0 — Re-baseline

v1's `perf/perf.js` ran 6 pages on Railway with 2 services. We now have ~20 routable surfaces on Fly with 4 services and a totally different realtime stack.

0.1 Synthetic baseline (Playwright) *(done — 2026-05-02)*

Shipped in `perf/perf-v2.js`. First baseline captured against staging on 2026-05-02 — 130 measurements (13 routes × 2 device profiles × 5 runs) written to `perf/baselines/perf-staging-2026-05-02T*.json`. Analysis in `doc/Performance_Snapshot_2026-05-02.md`.

- ☒ Extend `perf/perf.js` to cover the full route inventory.
- ☒ Two device profiles: desktop (1280×800, broadband) + mobile (Moto G4 / 4G via CDP throttling).
- ☒ Run against `xo-*-staging.fly.dev`. 5 runs each, p50 + p95.
- ☒ Capture: TTFB, FCP, LCP, Ready, JS bytes, image bytes, requests.
- ☒ Persist results into `perf/baselines/<env>-<ts>.json`.
- ☐ **Open follow-ups** — these are real gaps, *not* blockers for Phase 1; tackle when their inputs become available:
  - ☐ Wire `cold-signed-in` and `warm-signed-in` contexts (needs `TEST_USER_EMAIL` + `TEST_USER_PASSWORD` in env).
  - ☐ Run against `xo-*-prod.fly.dev` once Phase 3.8 is promoted.
  - ☐ Re-run on staging once it bumps past `1.3.0-alpha-8.0` so Sprint 3.8.B + 3.8.C are reflected.
  - ☐ Add INP measurement (one scripted interaction per page — open modal / click row / submit form).
  - ☐ Disable Fly auto-suspend on staging for the run, seed the four representative-data fixtures from the Pre-flight section.

0.2 RUM (real-user monitoring) *(not started — Tier 1)*

- ☐ Wire `web-vitals` into the landing app — POST `{ metric, value, route, deviceClass }` to a new `POST /api/v1/rum` endpoint with sampling (10% in prod, 100% in staging).
- ☐ Server stores into `RumSample` table (route, metric, value, percentile bucket, timestamp, country/region). Add Prisma model + migration.
- ☐ Admin dashboard panel: per-route p50/p75/p95 over 24h / 7d. (Reuse the existing admin metrics scaffold.)
- ☐ Define alert: any route's p75 Ready > 1.5× target for 30min → Slack.

0.3 Backend RED metrics (Rate / Errors / Duration) *(not started — Tier 2)*

- ☐ Add request-duration histogram middleware in `backend/src/index.js` and `tournament/src/index.js`. Bucket per route + status.
- ☐ Expose `/metrics` (Prometheus text format) — Fly already supports scraping.
- ☐ Backfill the Observability dashboard (`doc/Observability_Plan.md`) with p95 panels per endpoint family.

- ☐ Confirm SSE channel join + dispatch latency are instrumented (the realtime postmortem mentions per-channel timing — verify it lights up Grafana).

0.4 Targeted gap measurements (2026-05-04)

The 0.1 baseline left four open questions. New scripts plug each one without waiting on RUM (0.2) or Prometheus (0.3):

- **`perf/perf-inp.js`** — INP per page via `PerformanceObserver({type: 'event', durationThreshold: 16})`, 5 runs × desktop+mobile.
- **`perf/perf-sse-rtt.js`** — SSE connect, POST move ack, POST move → SSE state event, POST move → bot move event, 20 runs.
- **`perf/perf-backend-p95.js`** — concurrent loader, 200 reqs × concurrency 5 across the five hot read endpoints.
- **`perf-v2.js --extended-resources`** — wait 5s post-Ready and re-collect resource bytes, exposing late-loading images that `transferSize` reports as 0 at Ready time.

Saved at `perf/baselines/{inp,sse-rtt,backend-p95,perf}-{env}-*.json`. Headlines below; raw numbers in the JSON.

INP — the click-to-paint floor is excellent Where the script could find a stable interaction, every measured route sits **well under the 200ms "good" threshold** on both desktop and Moto G4 mobile:

Interaction	Desktop p50	Desktop p95	Mobile p50	Mobile p95
Home — refresh demo	**24ms**	120ms	24ms	32ms
Home — open sign-in	**32ms**	32ms	24ms	40ms
Puzzles — first button	**24ms**	24ms	24ms	24ms

Two routes (Tournaments filter, Leaderboard "Show bots" toggle) hit selector-miss issues during the run; the data we have already says **INP is not a problem on the platform today** for the routes that landed. Phase 15 (Guide INP audit) drops in priority — the *idle* floor is fine. The remaining concern is INP under live SSE updates, which the current script doesn't measure (TODO: extend with a "during-cup" mode).

SSE round-trip — the real perceived-latency story Every PvP / PvB move passes through this path. Numbers are the headline finding of this whole section:

Phase	Staging p50	Staging p95	Prod p50	Prod p95
SSE connect → session	164ms	567ms	207ms	304ms
POST move → ack	174ms	563ms	262ms	432ms
POST move → SSE state	**560ms**	838ms	**670ms**	1174ms
POST move → bot move event	560ms	838ms	670ms	1175ms

Three load-bearing facts here:

1. **The POST acks at 174–262ms but the SSE event arrives 386–408ms later.** That gap is the SSE pub/sub dispatch — flushing the channel write through to the same client's open EventSource. Players click a square and wait ~half a second to see the result. This is *the* perceived-perf bottleneck on every move, every game, every user. **Phase 5 promotes from Tier 2 to Tier 0.**
2. **Bot move and player move arrive ~simultaneously** because the backend dispatches both state events in the same request handler. Bot computation is effectively free. No work needed there.
3. **Prod is ~120ms slower than staging on the SSE round-trip.** Same machine class; same code. Likely cold-machine flap on the move POST handler — Phase 6 input.

Backend endpoint p95 — healthy across the board

Endpoint	Staging p50	Staging p95	Staging p99	Prod p50	Prod p95	Prod p99
`GET /api/version`	50ms	131ms	237ms	60ms	73ms	153ms

Endpoint	Staging p50	Staging p95	Staging p99	Prod p50	Prod p95	Prod p99
`GET /api/v1/bots?gameId=xo`	71ms	153ms	224ms	133ms	173ms	242ms
`GET /api/v1/leaderboard?game=xo`	52ms	114ms	121ms	62ms	85ms	97ms
`GET /api/auth/get-session`	57ms	140ms	158ms	62ms	74ms	93ms
`GET /api/tournaments`	66ms	142ms	166ms	85ms	107ms	166ms

Implications:

- **Phase 2 (DB index pass) is genuinely off the critical path for now.** No endpoint's p95 is over 175ms; p99 caps at 242ms (cold-bot list on prod). Promote only when a future feature plants a slow query.
- **Prod auth endpoint rate-limited at concurrency 5.** 110 of 200 requests returned 429 — Better Auth's default rate limiter is kicking in hard. Either bump the limit or whitelist the smoke runner. Either way: real users won't hit this, but admin scripts will.
- **`/api/v1/bots` is the slowest endpoint** (133ms p50 prod). Every cold-anon page that calls `getCommunityBot()` waits on this. If Phase 1 cuts page-level work, this becomes the next visible blocker — candidate for edge KV (Section E4) since the built-in bot list rarely changes.

Extended-resource capture — the hero image was never zero KB The 2026-05-02 snapshot reported `img_kb: 0` on every route. With `--extended-resources` (5s post-Ready wait), `colosseum-bg.jpg` shows up at **909 KB** on every measured route, starting around 500–700ms (overlapping the Ready window):

Route	Image start (ms)	Bytes
Home	534	909 KB
Leaderboard	676	909 KB
Tournaments	507	909 KB

Original perf-v2 was polling `transferSize` while the resource was still in flight, which Resource Timing reports as 0 until the body fully lands. **Phase 3 (image diet) promotes from Tier 1 to Tier 0**: 909 KB shipped on every page is the second-biggest single-byte cost after the JS bundle, and unlike JS it's reachable with one PR (AVIF + responsive sizes + skip-on-mobile).

Deliverable: a single `Performance_Snapshot_<date>.md` checked in next to this plan, showing where every route currently sits vs the targets above. *Every subsequent phase compares against this baseline.*

Section A — Real performance

Phases that move the actual latency floor (server time, network, parse, render). Ordered by expected impact-to-effort.

Phase 1 — Bundle audit + per-route splitting *(Tier 0 — start here)*

The 2026-05-02 visualizer pass (run `VISUALIZE=1 npx vite build` in `landing/`) showed `main.supported-*.js` at **1,529 KB raw / 411 KB gz** holding the entire app. Only `vendor-react`, `game-xo`, `game-pong`, and the Gym sub-tabs are split out — every page component is in `main`. Per- route splitting is the highest-leverage fix in the entire plan; until this lands, every other phase is a rounding error.

Concrete sub-steps (in order):

- ☐ **Per-route `React.lazy()` for every page in `App.jsx`.** PlayPage, GymPage, TournamentDetailPage, TournamentsPage, ProfilePage, BotProfilePage, PublicProfilePage, RankingsPage, StatsPage, SettingsPage, TablesPage, TableDetailPage, SparPage, PuzzlePage, plus all `/admin/*` pages. Wrap each `<Suspense fallback={<Skeleton/>}>` so the skeleton already aligns with Phase 8 work.
- ☐ **Vendor-split `@xo-arena/*`** in `manualChunks` — `xo`, `nav`, `ai`, `sdk`. They're currently bundled into `main.supported`. Promote each to its own chunk.
- ☐ **Audit Better Auth client.** Likely a meaningful chunk (auth-client + better-fetch + zod). If the auth flow is only relevant after a sign-in trigger, lazy-load the client module.

- ☐ **Audit `recharts`.** Only used in Gym Analytics + Stats. Confirm `vendor-charts` chunk separation; if it's leaking into `main`, fix the import boundary.
- ☐ **Audit ML engines (`@xo-arena/ai`).** Q-learning / DQN / AlphaZero should only load on Gym Train tab — already lazy at the tab level, but verify the engines aren't statically imported elsewhere.
- ☐ **Audit images / SVG / icons.** Anything imported as a JS module should be inspected; large inline SVGs should move to `<img>` or a sprite sheet (currently inline JSX SVGs ship inside the JS chunk).
- ☐ **Tactical library swaps.** Each of these has a measured-bytes payoff; do them as separate PRs so each can be validated:
  - `recharts` → `uplot` (~20× smaller, only used in Gym Analytics + Stats).
  - Audit `dayjs` boundary; nothing should pull `moment`, `luxon`, etc.
  - Subset `lodash` imports (`lodash-es/get`, not `lodash`); remove `lodash` if any path still uses it.
  - Audit `zod` boundary — only use on validators that actually run client-side.
- ☐ **Tree-shake hygiene.** Mark `sideEffects: false` on every internal `@xo-arena/*` package so unused exports drop. Audit duplicate deps (`npm ls react`, `npm ls zustand`).
- ☐ **Brotli at the edge.** Confirm Fly is serving Brotli, not just gzip — usually 15–20% smaller wire bytes for free.
- ☐ **Aggressive minifier.** Compare `esbuild` vs `swc` vs `terser` passes on `main.supported`. Cumulative ~5–10% gain typical.
- ☐ *(Optional / longshot)* Preact compat alias — saves ~50 KB gz over React 18, but Better Auth + react-router compat needs verification. Land last, behind a feature flag.
- ☐ Generate updated visualizer report; commit at `perf/baselines/bundle-<date>.html` for diffing.
- ☐ Re-run `perf-v2.js` against staging — confirm `main` < 200 KB gz, per-route chunks 30–80 KB gz, and FCP / Ready drop on every page.
- ☐ **CI bundle-size guard.** Fail PR if any chunk grows > 5% over `main` without a `perf-ok` label. (`size-limit` or hand-rolled.)

Expected outcome (data-driven hypothesis): - `main` chunk: 411 KB gz → ≤ 200 KB gz - First-paint JS: 493 KB gz → ~280 KB gz on Home, lower per-route - Mobile FCP: 1360ms → ~700ms (parse cost cut roughly proportional to byte cut on Moto G4 CPU) - Desktop Ready: 760ms → ~400ms

If the second perf-v2 run does *not* show these moves, Phase 1 is *landed but ineffective* and we go back to the visualizer to find what else is stuck in `main`.

Phase 1b — PlayVsBot deep-dive *(Tier 0, immediate after Phase 1)*

The 10-run confirmation showed `/play?action=vs-community-bot` is 600–700ms slower than the next-worst route on both devices, *every time*. The path is sequential and visible to a returning user every time they click "Play vs Bot" cold:

```
mount → getCommunityBot() fetch → /api/v1/rt/tables POST → redirect → render
```

- ☐ Trace each step server-side and client-side; identify the longest synchronous wait.
- ☐ Render an interim board *shell* (skeleton board + "Finding bot…") *before* `getCommunityBot()` returns — the spinner detection then flips when the bot is ready, not 600ms later.
- ☐ Move `getCommunityBot()` cache warming into the route entry — hover-prefetch `/play` from the home CTA, fire the bot fetch then.
- ☐ Verify `/api/v1/rt/tables` POST is the actual path (not socket). Make sure the server returns the table id immediately and SSE backfill picks up state, instead of a single big response.
- ☐ Re-measure with the same `--routes=PlayVsBot --runs=10` invocation. Target: parity with `/play` (≤ 750ms desktop, ≤ 1600ms mobile).

Phase 2 — Database query audit + indexing pass

The schema has grown a lot since v1. Audit slow queries across hot endpoints, especially the new ones:

- ☐ Enable Postgres `pg_stat_statements`; collect 24h of staging traffic.
- ☐ Top 20 by total time + top 20 by mean time — review each.
- ☐ Critical paths to check: tournament list (`/api/tournaments`), tournament detail with rounds + participants + matches eager-loaded, leaderboard, bot list with skills, ML session list, table presence lookups.
- ☐ Add missing indexes (most likely: `TournamentMatch (tournamentId, roundId)`, `TournamentParticipant (tournamentId, status)`, `BotSkill (botId, gameId)` — verify the partial unique still covers reads).
- ☐ Convert any lingering N+1s to `findMany` + grouped indexing (`groupBy`/`include`).
- ☐ **Materialized view for the leaderboard.** The current SQL aggregates win counts on every request. Refresh the view on game-end (already a single hot path). Read becomes O(rows returned), write cost moves out of the request thread.

- ☐ **Read replica for hot reads** (Fly Postgres supports it). Public read endpoints (`/leaderboard`, `/bots`, `/tournaments`) go to the replica; writes + auth stay on primary. Cuts write- lock contention on busy windows.
- ☐ **Stream large responses.** `GET /api/tournaments/:id` currently returns the full eager tree as one payload. With Express `res.write()` chunks (or NDJSON), the client can paint the header → bracket → participant table → match history in stages instead of waiting on the longest one.
- ☐ **Worker queue for off-request work.** ML training kicks, ELO calibration, journey event publishing — every one of these blocks the request that triggered it today. Push to BullMQ-style queue (Redis-backed); the request returns immediately.
- ☐ **Pre-warm critical caches on boot, not first request.** Built-in bot roster, puzzles, system config — load on `app.listen` callback so the first user doesn't pay the miss.
- ☐ **Compress payload responses.** Confirm gzip/Brotli on JSON responses, not just static assets — cuts wire bytes 60–80% for tournament/leaderboard JSON.

Phase 3 — Hero image + asset diet ***(Tier 1 — confirm impact first)***

The 2026-05-02 baseline showed **0 KB image bytes** on every route — the `colosseum-bg.jpg` (888 KB) didn't appear in the static byte total. It likely loads async after Ready resolves, or via a CSS background not caught by the resource-type filter. Before sinking time into AVIF encoding, confirm where and when the image actually lands.

- ☐ **Investigate first.** Open DevTools → Network on `/` cold, filter by Img. Confirm whether `colosseum-bg.jpg` loads at all on cold- anon, when it loads relative to FCP / Ready, and on which routes.
- ☐ If it loads after Ready: low-priority — it doesn't move the measured budget. Still worth shipping AVIF + responsive sizes for mobile data costs, but demote out of Tier 1.
- ☐ If it blocks LCP on `/`: keep in Tier 1 — convert to AVIF + WebP + JPEG fallback via `<picture>`; cap mobile delivery at ≤ 1280 wide; preload.
- ☐ Audit other public images (`landing/public/`); drop anything unused.
- ☐ Add `loading="lazy"` and explicit `width`/`height` on every `<img>` to stop layout shift (CLS).
- ☐ Confirm font strategy: `font-display: swap` and self-hosted (no blocking Google fonts).
- ☐ **Subset fonts** to the glyphs actually used. A full Latin Inter subset is ~25 KB; full font is ~120 KB.
- ☐ **SVG sprite sheet** for icons. Inline JSX `<svg>` ships inside the JS chunk — moving icons to an external sprite (`<use href />`) lets them stream + cache separately and shrinks `main.supported`.
- ☐ **Skip the hero on mobile entirely.** A CSS gradient or solid colour as fallback; serve the real image only at ≥ 1024 viewport. Cuts ~200 KB off mobile data on `/`.

Phase 4 — Cross-service network shape

The landing server proxies `/api/(classification|recurring|tournaments)/*` to the tournament service. Each proxied call adds a hop.

- ☐ Measure: cross-service p50/p95 vs same-service.
- ☐ If the gap is meaningful, consider:
  - Direct browser → tournament service via a `tournament.<env>.fly.dev` subdomain + CORS (saves the proxy hop).
  - Or, push read-mostly endpoints (recurring list) into the backend with Redis-backed sharing.
- ☐ Audit `Connection: keep-alive` between landing↔backend, landing↔tournament — don't pay TLS handshake on every request.
- ☐ Confirm **HTTP/3 (QUIC)** is enabled on all Fly fronts. Cuts handshake latency on flaky mobile connections (vs HTTP/2's head-of-line blocking).
- ☐ **Same-origin auth cookie path.** Better Auth sometimes triggers a CORS preflight `OPTIONS` for `/api/auth/get-session`. If the cookie domain is set tight, this is `Access-Control-Max-Age` away from a noticeable savings (or eliminate the preflight entirely).
- ☐ **Audit cookie size.** Better Auth tokens can balloon (esp. with provider claims). Every request carries them — keep < 4 KB.
- ☐ **Combine cold-start API calls into `/api/v1/init`.** A returning signed-in user fires session check + bots + tournaments + journey progress in parallel; one combined endpoint lets the server fan out + return one payload, cutting N waterfalls to 1.

Phase 5 — SSE channel + POST round-trip latency

The realtime layer was rewritten between v1 and v2 (see `doc/Realtime_Migration_Postmortem.md`). It needs its own performance lens:

- ☐ Measure: `EventSource` open → first event latency; channel join → server ack; client POST move → server ack → SSE state event arrival.
- ☐ Look for redundant subscriptions in `useEventStream` (one channel per tab vs one shared singleton — the postmortem mentioned a singleton fix; verify it's still in place).
- ☐ Move idle/heartbeat traffic to a long-poll or coalesce into one channel to keep open connections low (Fly per-instance HTTP limits).
- ☐ Backpressure: under load, what happens when one slow consumer holds an SSE write? Add a watchdog.

Phase 6 — Backend cold start / warm path

Fly machines suspend on idle. First-request latency after suspend can dominate Ready for low-traffic regions.

- ☐ Measure cold-start time per service (backend, tournament, landing).
- ☐ Decide per service: `auto_stop_machines = false` for hot services (backend), keep for cold (admin?). Document the cost trade-off.
- ☐ Pre-warm critical caches (built-in bots, puzzles) on boot, not on first request.
- ☐ Move synchronous init out of the request path (Prisma engine load, journey config load).

Phase 7 — Mobile-specific

Mobile is now first-class but never benchmarked. The 2026-05-02 snapshot showed Mobile FCP at ~1360ms — that's the parse cost on a Moto G4 CPU. After Phase 1 cuts the bundle, the remaining mobile gap will come from critical-path render and CSS.

- ☐ Identify any desktop-only assumption: large tables, hover-only UI, keyboard shortcuts, missed touch targets.
- ☐ Audit DOM size on the heaviest mobile pages (TournamentDetailPage, GymPage with tabs).
- ☐ Lighthouse mobile score ≥ 90 per route as a hard CI gate.
- ☐ **Critical CSS inlined into `<head>`.** Most CSS is render-blocking today (`<link>` tag → server round-trip on bad networks). Extract the above-the-fold subset and inline it; defer the rest. Likely saves 50–150ms on Moto G4 / 4G FCP.
- ☐ **`touch-action: manipulation`** on tappable areas. Eliminates any residual 300ms tap delay; cheap one-liner.
- ☐ **Transform-only animations.** Audit any animation that uses `top`/`left`/`width`/`height` — replace with `transform` so it runs on the compositor thread, not main.
- ☐ **`content-visibility: auto`** on long off-screen sections (the tournament participant table when collapsed, the leaderboard rows below the fold, the Gym session list). Free skip-render for non-visible content; one CSS line per container.
- ☐ **Reduce DOM nodes** on mobile-heavy pages. TournamentDetailPage's 1900 lines render hundreds of nodes — split via Suspense and only mount visible sections.
- ☐ Service Worker for offline-first shell (covered in Phase 21 below).

Section B — Perceived performance

These don't move the latency floor but make the platform *feel* faster. v1 covered skeletons, hover prefetch, optimistic moves, optimistic writes — all good and shipped. v2 extends to the new surfaces.

Phase 8 — Skeletons everywhere new

Every page added since v1 should have a content-shaped skeleton, not a spinner.

- ☐ `TournamentsPage` — list of tournament cards skeletons.
- ☐ `TournamentDetailPage` — header + bracket + participant table skeletons.
- ☐ `BotProfilePage` — bot header + skill cards + ELO chart.
- ☐ `Profile` — accordion frames render with skeletons inside.
- ☐ `Spar` / `Tables` — table card skeletons.
- ☐ Audit: any place where a `<Spinner />` is the first thing rendered after route change is a skeleton candidate.

Phase 9 — Optimistic / streaming everywhere

- ☐ Tournament register: optimistic "Registered □" pill before the server ack (with rollback on NO_SKILL etc.).
- ☐ Bot create: bot card appears immediately with a faded "saving…" badge.
- ☐ Skill add: pill appears in `setBots` optimistic merge (already done in Profile — verify Gym uses the same path).

- ☐ Move events: already optimistic for player; verify bot moves animate smoothly when the SSE event arrives.
- ☐ Streaming responses for large GETs (tournament detail with 100+ matches) — render the bracket as JSON streams in.
- ☐ **Partial-UI loading states.** Replace `Loading…` with progress narration: "Found bot, building table…" / "Bracket ready, fetching participants…". Same wait time, lower perceived latency.
- ☐ **Animation cover during fetch.** A 200ms fade/slide on route change can hide a 200ms fetch entirely — the wait reads as polish.

Phase 10 — Hover-intent prefetch + "warm next" patterns

v1 added hover prefetch on nav links. Extend:

- ☐ On `/tournaments`, hover-prefetch each tournament's detail JSON.
- ☐ On `/profile?section=bots`, hover-prefetch `/gym?bot=:id` chunks for each row.
- ☐ On match end (post-cup), prefetch the next likely route (next match, coaching card).
- ☐ On `/play`, after first move, prefetch the rematch flow's chunks.
- ☐ **Pre-render the next likely route** in a hidden iframe / shadow DOM during idle time on slow paths (post-cup celebrations, match results). Click feels instant.

Phase 11 — Route transition polish

- ☐ Re-evaluate the v1 fade transition under React 18 Suspense — does it conflict with streaming UI?
- ☐ Add `view-transitions` API where supported (Chrome) for snap-zero route transitions on the same-document navigations.

Phase 12 — In-game feel

- ☐ Confirm sound latency is still nailed (the capture-phase pointerdown listener in `soundStore.js` — per CLAUDE.md, must not be removed).
- ☐ Animation budgets: every move, popup, modal — none > 200ms.
- ☐ Reduce-motion respect on every animation.

Section C — Newly heavy surfaces (deserve their own pass)

Phase 13 — Tournament page perf

`TournamentDetailPage` is the heaviest page in the app (1900+ lines, eager loads rounds + participants + matches + their users). It runs before / during / after the cup journey and shows under load on Cup days.

- ☐ Hard target: cold mobile Ready ≤ 600ms even with 32 participants and a full bracket.
- ☐ Split the page: bracket + participant table + match history are independent, render each behind its own Suspense boundary.
- ☐ Server: paginate completed matches; only the current round needs full detail.
- ☐ Memoize bracket layout math; profile re-renders when SSE events fire.
- ☐ Confirm `ParticipantTable` disambiguation (just shipped in 3.8.C) doesn't do per-render allocation.
- ☐ **`useDeferredValue`** on the bracket + participant lists when SSE pushes a new event. Render the new state at lower priority than user input (React 18).
- ☐ **Memoize `BracketMatch`** — currently re-renders on every SSE event for the whole bracket, even matches that didn't change.
- ☐ **`content-visibility: auto`** on rounds the user has scrolled past or off-screen.

Phase 14 — Gym page perf + ML compute

The Gym detail panel re-mounts whole tab trees on bot/skill switch, and ML inference today runs on the main thread (jank during play).

- ☐ Ensure keep-alive (`display: none`) survives the new bot→skill drilldown.
- ☐ ML model fetch should stream — show training history first, then current model weights.
- ☐ **Web Worker for ML inference.** Offload `getMoveForModel` for ML bots to a worker so the main thread never janks during a move.

- ☐ **WASM minimax / AlphaZero.** Could move bot move latency from ~30ms to <5ms. Compile the minimax engine to WASM behind a flag; benchmark vs JS. Bigger payoff once Connect4 lands (state space grows).
- ☐ **OffscreenCanvas for game render** (where applicable). Frees main thread for UI during high-FPS games like Pong.
- ☐ **`requestIdleCallback`** for journey events, telemetry, and RUM flushes — never compete with user input.

Phase 15 — Intelligent Guide journey

The journey adds bus events, popups, scrim spotlights, and re-renders on every step trigger. Measure — it's a likely INP regressor.

- ☐ INP audit on each step trigger; confirm < 100ms.
- ☐ Memoize `JourneyCard` re-renders.
- ☐ Lazy-load step-specific components (TrainGuidedModal, BotCreatedPopup).
- ☐ Confirm guide store mutations don't cascade through the whole tree (split selectors).

Phase 16 — Tables-as-primitive overhead

Tables underlie spar, journey demos, PvB, PvP, and tournament HvB. Verify the common path is hot:

- ☐ `tablePresence` lookups: O(1) per join? Memory footprint at 1000 active tables?
- ☐ `botGameRunner`: parallel runs OK at 50 concurrent demo tables?
- ☐ `createTableTracked` resource counter overhead — single Redis call or multi-hop?

Section C2 — Compute, cache, and architecture (added from brainstorm)

These are bigger swings than the per-phase tactics above. Each one is a "could be a project" rather than a checklist item, but they're listed here so the option is visible when the data points to them.

Phase 19 — Compute relocation

Today, all heavy compute runs on the user's main thread:

- ML training (Q-learning, DQN, AlphaZero) — browser-only.
- ML inference (`getMoveForModel`) — main thread.
- Game logic (minimax, AlphaZero MCTS) — main thread JS.
- Journey events, ELO calibration — request thread on backend.

Targets after this phase:

- ☐ **Web Worker for ML inference.** Offload to a worker; main thread never janks during a move. Pairs with Phase 14.
- ☐ **WASM for minimax / AlphaZero.** Compile the engines to WASM behind a feature flag; benchmark vs the JS engines. Expected 10–100× faster move computation. Critical for Connect4 (tied to `doc/Connect4_Ship_Checklist.md` Spike A on TF.js).
- ☐ **OffscreenCanvas for game render.** When games like Pong land, keep the render loop off the main thread.
- ☐ **Move ML training to a backend worker.** Tracked in `doc/Connect4_Ship_Checklist.md` Spike B — perf is one input; architectural decision is broader (mobile training, headless cluster runs, training on devices that can't keep a tab open).
- ☐ **`requestIdleCallback`** for journey events, telemetry, RUM flushes — never compete with user input on the main thread.

Phase 20 — Cache & app shell

Today: warm visits read from `localStorage` for some stores, but the HTML / JS / CSS still hits the network every time.

- ☐ **Service Worker app shell.** Cache the index HTML + the vendor chunks; serve from cache on second visit; revalidate in background. Repeat-visit Ready ≈ 0ms. Highest-leverage perceived-perf change not yet in v2.
- ☐ **IndexedDB for tournament/leaderboard JSON.** Faster than localStorage at size; survives reload. Stale-while-revalidate directly off IDB.
- ☐ **Edge response cache** on Fly for public reads (`/leaderboard`, `/bots`, `/tournaments`). Backend never sees cached requests.
- ☐ **HTTP `immutable` cache headers** on hashed asset URLs (`max-age=31536000, immutable`). Confirm Fly is doing this.
- ☐ **Persist Zustand stores** so warm boot has full state — guide progress, sound prefs, ui sort orders, etc.

Phase 21 — Architecture experiments (longshots)

Each is a meaningful re-architecture; pursue only if Tier 0 + Tier 1 data shows the gap they'd close. Listed for completeness.

- ☐ **Server-side render the landing page (`/`) only.** It's mostly static. React 19 RSC or a tiny SSR shell could deliver Ready ≈ TTFB. Other routes stay SPA.
- ☐ **Edge functions for `/leaderboard` / `/bots`.** Fly Functions could serve these from cache without touching the Node process, cutting cold-start exposure entirely.
- ☐ **Replace JSON-over-HTTP with gRPC-Web** for hot endpoints — smaller payloads, schema validation, generated clients.
- ☐ **Migrate hot read endpoints off Express to Hono on the edge** — lower per-request overhead than full Express middleware stack.
- ☐ **React Server Components for static-ish pages** — Settings, Puzzles, parts of Profile. Cut their JS to ~0 KB.

Section D — Instrumentation as a first-class deliverable

Performance work without instrumentation is fishing in the dark. v1 measured synthetically once and stopped. v2 makes measurement continuous.

Phase 17 — Per-PR perf gates

- ☐ CI runs `perf/perf.js` against a preview env on every PR; comments the diff vs `main`.
- ☐ Bundle-size guard fails PR if any chunk grows > 5% without an explicit `perf-ok` label.
- ☐ Backend test suite asserts p95 < budget for the 5 hottest endpoints (run against an in-memory load generator).

Phase 18 — Production perf dashboard

- ☐ Public-internal Grafana page: per-route Ready p75/p95 (RUM), per-endpoint backend p95, SSE channel health, error budgets.
- ☐ Weekly "perf review" cadence — owner reviews against budgets, files issues for any breach > 7 days.

Section E — Aggressive bets (re-review 2026-05-04)

A second pass over the plan exposed several large-leverage ideas that weren't in any phase. Some are nearly-free additions to existing phases; others are real moonshots. They're grouped by what they buy you, not by tier — surface them where they fit, then promote the ones the data backs after Tier 0 lands.

E1 — Free wins (add to Tier 0 / quick-wins)

These cost ≤ a day each, ship behind feature flags if needed, and should be done alongside the bundle work — most don't even require a benchmark to justify.

- ☐ **103 Early Hints from Fly.** Send `<link rel="preload">` for `vendor-react` + `main` + critical CSS *before* the backend has finished computing the HTML response. Saves 50–150ms by overlapping TTFB with preload. Fly's edge supports this; verify and wire it up.
- ☐ **Speculation Rules API** (`<script type="speculationrules">`). Chrome / Edge will prerender same-origin links the user is likely to click — no JS, no library. Add a JSON block with the home, `/play`, `/leaderboard`, and `/profile` URLs. Free instant navigation on Chromium.
- ☐ **HTTP/3 0-RTT for returning visitors.** Cuts the TLS handshake to zero on warm connections. Fly supports HTTP/3; confirm 0-RTT isn't disabled for safety reasons we don't actually need.
- ☐ **Resource hints for auth providers** — `<link rel="preconnect">` / `dns-prefetch` for `accounts.google.com` and `appleid.apple.com` in `<head>` so the OAuth round-trip starts warm. Sign-in feels ~200ms faster on cold visitors who click immediately.
- ☐ **Brotli quality 11 at build** (vs Vite's default 6). Slower build, zero runtime cost, ~5–10% smaller JS payloads. Set `compress: { algorithm: 'brotliCompress', compressionOptions: { level: 11 } }` in the Vite config.

- ☐ **CSS `image-set()`** for the hero — `background-image: image-set("hero.avif" 1x type("image/avif"), "hero.webp" 1x)`. Browser picks best format automatically; supersedes manual `<picture>` wrapping for CSS backgrounds.
- ☐ **CSS `aspect-ratio`** on every `<img>` and lazy section. Locks layout before bytes arrive — kills CLS on the home, leaderboard, and tournament pages without manual `width` / `height`.
- ☐ **BlurHash placeholders** for the hero, bot avatars, and any future user uploads. ~30 bytes per image, decoded synchronously to a 32×32 canvas — instant visual content under the real image.
- ☐ **Variable fonts.** Replace separate Inter regular/semibold/bold static files with one variable font (`Inter.var.woff2`). Roughly half the font weight on the wire.
- ☐ **`font-display: optional`** for non-critical text (e.g., Inter Tight on hero headlines). Never blocks render, even on slow networks.
- ☐ **Modern-only build by default.** Drop the legacy bundle in production (Vite still builds it). 99% of traffic is evergreen Chromium / Safari / Firefox; the legacy chunk is dead weight.
- ☐ **`scheduler.yield()`** in any loop > 5ms (bracket layout, journey derivations, leaderboard sort). Hands control back to the browser between chunks of work — keeps INP under 100ms even on Moto G4.
- ☐ **`isInputPending()`** as the cheap version when `scheduler.yield` isn't shipped: pause heavy work mid-loop if the user is touching the screen.

E2 — React re-render lockdown

The existing phases mention `useDeferredValue` and `React.memo` on `BracketMatch`, but the platform has a deeper render-cost story.

- ☐ **Atomize Zustand stores.** The current game store, journey store, and ui store each have many subscribers. Splitting each into per-domain slices (e.g., `tableStore`, `seatsStore`, `boardStore`) means an SSE state event re-renders only what actually changed. Easy diff pattern; large measurable win on TournamentDetail under live updates.
- ☐ **`useTransition` on every nav click.** Keeps the current page interactive while the next route's chunks load — visible Ready stays low even when the next page is heavy.
- ☐ **Selective hydration where applicable.** React 18 already supports it; needs explicit `<Suspense>` boundaries to opt in. Pair with Phase 1's per-route splitting.
- ☐ **Why Did You Render audit** on TournamentDetailPage, GymPage, RankingsPage. Run in dev for a session, fix every flagged cascade. One-time effort, perpetual savings.
- ☐ **React 19 Actions + `useFormState`** for sign-in, register, bot-create, skill-add. Cleaner optimistic UI and built-in pending state — replaces hand-rolled patterns.
- ☐ **React Forget (compiler memoization)** — track stability; adopt when the React team marks it production-ready. Removes most `useMemo` / `useCallback` boilerplate, often catching cases hand-written memoization missed.

E3 — Compute moonshots (Phase 14 / 19 supercharger)

The existing plan moves ML to a Web Worker. The real wins are bigger.

- ☐ **WebGPU backend for TF.js.** Newer than the WebGL backend, ~2–5× faster on supported hardware. Falls back to WebGL automatically. One config change in TF.js setup.
- ☐ **WASM SIMD for minimax.** Compile the minimax engine with the `simd` flag — 4–8× speedup over scalar WASM, which itself is ~10× faster than the JS version. Bot move latency goes from ~30ms to ~1ms; matters more for Connect4 + AlphaZero.
- ☐ **Quantize ML models to int8.** TensorFlow.js supports int8 quantization at the conversion step. 4× smaller weights on disk + on the wire, 2–4× faster inference on CPU. Trade is ~1–2% accuracy, which is well within ELO noise for our game.
- ☐ **Cache compiled WASM in IndexedDB.** WebAssembly.Module is structured-cloneable. Compile once, store, retrieve on next boot — saves the JIT pass on every reload.
- ☐ **Background-train ML bots while idle.** When the user is reading the home page or watching a demo, run a few training epochs in a Web Worker. Their bot quietly improves between sessions.
- ☐ **Speculative bot move computation.** During the user's turn, precompute the bot's response to the top 3–5 likely user moves. User makes their move → bot's response is already cached → animate immediately.

E4 — Network moonshots

- ☐ **PgBouncer in front of Fly Postgres.** Connection pooler that handles transient connection storms (deploy-time, cold-start flap). Cuts p95 tail latency on DB-bound endpoints. Standard Postgres infra; runs as a sidecar machine.
- ☐ **DataLoader for N+1 batching.** Especially on tournament-detail eager loads — fan out N participant lookups → one batched `SELECT … WHERE id IN (…)`. Battle-tested Node pattern; drop-in.
- ☐ **WebTransport (HTTP/3 datagrams) replacing SSE.** Lower latency than SSE, better connection migration when mobile users switch networks, supports unreliable channels for fire-and-forget. Big lift; do only after RUM data shows SSE is the bottleneck.
- ☐ **WebRTC peer-to-peer for PvP** once tables are seated. Server only relays signaling; gameplay traffic is browser-to-browser. Move latency drops to one round-trip on the same network. Big lift; meaningful only for PvP, not PvB.
- ☐ **Edge KV for read-heavy public reads.** Built-in bots, system config, journey config — load once at the edge, serve from memory; backend never sees these requests. Kills the request class entirely. Fly supports this via a small KV sidecar or Cloudflare Workers KV.
- ☐ **Stale-while-revalidate everywhere safe.** `Cache-Control: max-age=60, stale-while-revalidate=600` on `/leaderboard`, `/bots`, `/tournaments` list. Browser shows cached instantly, fetches fresh in background. One-line back-end change.
- ☐ **Compression dictionaries.** Brotli supports shared dictionaries across requests — train a dictionary on common JSON shapes (tournaments, bots, leaderboard) and ship it. 30–50% smaller on the wire after the first request. Browser support arriving 2026.

E5 — Cache & shell expansion (Phase 20 supercharger)

The existing Phase 20 mentions Service Worker app shell. Push it harder.

- ☐ **Service Worker as the canonical realtime substrate.** SW holds one shared SSE connection for all open tabs of the app — instead of N tabs × 1 SSE = N connections, it's 1 connection multiplexed via `BroadcastChannel`. Fly per-IP connection limits stop biting.
- ☐ **Background Sync for offline moves.** User makes a move on a flaky network → SW queues it → replays when connection returns. Player never sees a failed move.
- ☐ **IndexedDB for SSE event replay.** SW persists every state event to IDB; on tab reopen / reconnect, replay all missed events locally before the live SSE catches up. Reload feels instant.
- ☐ **Periodic Background Sync** for tournament wakeups. SW pings `/api/tournaments/upcoming` every 15 minutes when offline; pushes a notification when one starts. PWA primitive, free.
- ☐ **Push notifications for cup events.** Already partially wired via `pushSubscribe.js` — extend to "your match is starting" and "your bot won" pings.
- ☐ **WebShare API** on results pages so users share their cup wins to native iOS/Android share sheets. Not perf-direct, but cuts the friction loop that drives engagement → repeat sessions where the SW shell pays off.

E6 — Predictive perf

The platform has rich session data; use it to make navigation feel psychic.

- ☐ **Predictive prefetch from user history.** Track per-user navigation transitions (e.g., "users who view profile next view leaderboard 80% of time"). Prefetch the top-1 next route on every page load. Costs one tiny model + a few KB of state.
- ☐ **Pre-render the *expected* next route** in a hidden subtree. On `/play` cup match end, the next click is almost always "Continue" or "Rematch" — both routes can be fully mounted and ready when the click happens.
- ☐ **Speculative API calls based on cursor proximity.** As the cursor approaches a tournament card, fire `/api/tournaments/:id` before the click. Hover-prefetch as a behavior model, not just a CSS pseudo-class.

E7 — Architecture longshots (Tier 4)

Real re-architectures. Each is a project. Listed for completeness so the option is visible.

- ☐ **Bun runtime for backend + tournament services.** Cold-start drops from ~500ms (Node) to ~50ms (Bun) and request throughput goes up ~30%. Bun is Express-compatible; the migration is mostly testing. Collapses Phase 6 (cold-start) entirely. Risk: Prisma + Bun stability, Better Auth compatibility.
- ☐ **Hono + edge runtime for read-only public APIs.** Move `/leaderboard`, `/bots`, `/tournaments` list to Hono on Fly's edge. ~10× faster per request than Express, runs in every region. Already in Phase 21 — promote if Bun isn't ready.

- ☐ **Cloudflare Durable Objects (or Fly Replay) for tablePresence.** Move the in-memory presence cache to a regional state primitive. Frees backend machines to scale horizontally; eliminates a hot shared-memory table.
- ☐ **gRPC-Web for hot endpoints.** Schema-validated, smaller payloads, generated clients. Already in Phase 21; revisit after Bun.
- ☐ **Replace Better Auth with self-rolled JWT + refresh.** If the cookie-size + preflight overhead in Phase 4 measurements turns out to be load-bearing, the entire auth library may be the cheapest thing to swap. High-risk; only with measurement.

Section F — Pre-launch review gaps (2026-05-04)

A second pass before starting Tier 0 surfaced gaps the existing phases don't cover, plus DB / REST patterns the user explicitly asked about that the plan was silent on.

F1 — DB join shape + preloading (Phase 2 supplement)

Phase 2 calls for an index pass + `pg_stat_statements`, but doesn't audit the *join shape* of the heaviest endpoints. Hot offenders:

- **`GET /api/tournaments/:id`** — three-level `include`:

  ```
  include: {
    participants: { include: { user: { select: { …8 cols } } } },
    rounds:       { include: { matches: true } },
  }
  ```

  Prisma issues this as ~3 SQL queries with deep joins. On a 32-player bracket: 1 tournament + 32 participants joined to users + 5 rounds + 31 matches = a single response Of ~70 rows. Watch for two issues:

  (a) `matches: true` returns *all* TournamentMatch columns including large unused fields (e.g. `metadata` JSONB, `botMoveLog` if present);
  (b) the participant→user join can pull users that are no longer competitively active. *Action:* explicit `select` on every nested include, drop unused columns; benchmark before/after.

- **`listBots()` ELO lookup uses an OR over `(userId, gameId)` pairs.**

  ```
  where: { OR: eloPairs }   // N pairs → N OR branches
  ```

  Postgres can plan this poorly past ~8 branches. Rewrite as a single `WHERE (user_id, game_id) IN (VALUES …)` (raw SQL) or a UNION of exact-match subqueries; or add a composite `(user_id, game_id)` index if not already present. Cheap, measurable.

- **N+1 in spectate / live-state paths.** The bot-game runner emits state events that include only ids; the client backfills with separate fetches per id. Audit `useEventStream` consumers for any pattern that triggers a fetch loop on each event arrival.

- **Audit `select` discipline platform-wide.** Prisma defaults to selecting all scalar fields when `select` is omitted on a nested include. Three places in `listBots` already do this right; verify every `findMany`/`findUnique` in `backend/src/services/*` and `tournament/src/routes/*` follows suit.

F2 — REST response preloading (NEW)

The cold-page Ready cost is dominated by waterfalls: HTML parses → JS parses → bundle queries an API → API returns → render. The plan covers the JS parse but not the *API queries during parse*.

Instrumentation landed (2026-05-04, `dev`): `perf/perf-waterfall.js` measures, per route × device, when the primary `/api/*` call fires (`requestStart` from navStart) and when its response lands (`responseEnd`). Local sanity-check (n=3, desktop, no network throttling) — every route's primary API doesn't fire until ~400ms after navStart, which is pure JS parse + React mount + `useEffect` cost (the API itself is 2–5ms locally):

Route	apiStart	apiTotal	LCP	preload prize (~)
`/`	418ms	5ms	696ms	~400ms
`/tournaments`	377ms	2ms	388ms	~370ms
`/rankings`	385ms	5ms	396ms	~380ms

Route	apiStart	apiTotal	LCP	preload prize (~)

LCP on `/tournaments` and `/rankings` lands ~11ms after the API returns — preload would shift the entire chain ~370ms earlier. Staging/prod prize is presumably larger (longer bundle download + bigger API total). Re-run with `--target=staging` to confirm before committing implementation effort.

Aside: the waterfall also surfaces ~7 redundant `/api/session` hits per cold page load. Filed as a separate dedupe issue; not part of F2 itself.

- ☐ **`<link rel="preload" as="fetch">`** for the top-1 API call per route, emitted in the HTML `<head>` so the request fires in parallel with bundle download. Routes + targets:
  - `/` → `/api/v1/bots?gameId=xo` (the community bot fetch)
  - `/tournaments` → `/api/tournaments`
  - `/rankings` → `/api/v1/leaderboard?period=all&mode=all&includeBots=false`
  - `/profile` (auth) → `/api/v1/bots?ownerId=…`
- ☐ **103 Early Hints with `Link: rel=preload`** (Section E1) for both static chunks *and* the API preload above. Doubles the effect: preload starts before TTFB *and* parallel to JS parse.
- ☐ **Inline initial payload into `index.html`** for routes whose data shape doesn't depend on auth or query params. Static-ish candidates: built-in bot list, system config keys the client reads first paint. Cuts an entire round-trip on `/`.
- ☐ **HTTP/2 server push fallback?** Skip — browsers have removed it. 103 Early Hints is the modern equivalent and already in E1.
- ☐ **Wire `<Link to>` hover into a fetch warmup** (extending Phase 10): on hover, pre-fire the route's primary API call and cache the response for the upcoming nav.

F3 — Connection pool, transactions, query plans (NEW)

Phase 2 mentions PgBouncer; doesn't audit what's already in place.

- ☐ **Prisma connection pool size.** Default is `num_cpus * 2 + 1`, which on Fly's `shared-cpu-1x` resolves to 3. Under modest concurrency (5+ simultaneous SSE move POSTs) this saturates and queues. Bump explicitly via `connection_limit` query param; benchmark p95 under load before/after.
- ☐ **Transaction footprint audit.** Hot multi-step transactions: bot-game runner per-move, tournament round advancement, training-session completion. Long transactions hold row locks and block concurrent reads. Profile + shrink scope where possible.
- ☐ **Prepared statement reuse.** Prisma's query engine prepares and caches statements per connection — confirm the cache is sized for the actual query variety (default 100; if exceeded, Prisma re-prepares on every call).
- ☐ **`EXPLAIN ANALYZE` the top 5 slow queries** identified in the Phase 2 `pg_stat_statements` pass. A confirmed-slow query plan is what justifies an index, not a heuristic.
- ☐ **`SELECT FOR UPDATE` audit.** `tournamentService` advancement uses row locks; verify the lock window is bounded (no awaiting external service calls inside a `$transaction`).

F4 — Phase 5 (SSE round-trip) decomposition (concrete steps)

The 0.4 data shows POST move → SSE event = 560ms / 670ms p50, with ~400ms unaccounted for between POST ack and event arrival. The existing Phase 5 lists this at a high level; here's the specific sequence.

- ☒ **Time-trace one move end to end.** *(spike landed 2026-05-04 on dev)* — `backend/src/routes/realtime.js` POST `/rt/tables/:slug/move` emits `Server-Timing: lookup;dur=X, apply;dur=Y` (lookup = caller resolution + table findFirst; apply = applyMove + DB writes + Redis XADD). `backend/src/lib/sseBroker.js#dispatchEntry` injects `_t: { publishToPickupMs, pickupToWriteMs }` into the JSON payload of every SSE frame (publishMs derived from the Redis stream id `<ms>-<seq>`). `perf/perf-sse-rtt.js` parses both and reports six decomposition rows: server.lookup, server.apply, network (POST RTT − server), redis publish→pickup, broker pickup→write, network (ack→event − server). **Local sanity-check (n=10):** every leg ≤2ms; total matches client-measured `playerEventMs`. Staging + prod numbers TBD after promote — that's where the ~400ms lives.
- ☐ **Confirm SSE response is unbuffered.** Express + compression middleware can buffer SSE frames if not disabled per-route. Verify `Content-Encoding: identity` (or no-op compress) on `/api/v1/events/stream` — gzip would buffer the entire stream.

- ☐ **Disable Nagle's algorithm** on the SSE response socket (`socket.setNoDelay(true)`). On Node + Express this is rarely automatic; can shave 40ms per write.
- ☐ **Audit Redis pub/sub round-trip.** SSE channels go through Redis (per `useEventStream`). Time the publish→deliver gap on the same VM. If it's > 50ms, the pubsub topology needs work (sharding, in-process fanout for same-instance subscribers).
- ☐ **Same-instance shortcut.** If the publishing process and the SSE-serving process are the same Fly machine (true for our single-machine setup), skip Redis entirely and dispatch via in-memory event emitter. Redis only kicks in when machines need to fan out to subscribers on other VMs.
- ☐ **Coalesce duplicate events.** A single move can emit `state(moved)` + `lifecycle(...)` + journey progress; if these fire serially with their own SSE writes, latency multiplies. Pack into one frame where possible.
- ☐ Re-measure with `perf-sse-rtt.js`. Target: POST move → SSE event ≤ 200ms p50 staging, ≤ 250ms p50 prod.

F5 — Phase 1 specifics from the visualizer (concrete byte targets)

Phase 1 gives directional sub-steps but no per-chunk byte budget. After running `VISUALIZE=1 npx vite build` once and reading the output, set hard targets for each chunk and fail CI on regression:

- ☐ `vendor-react` ≤ 50 KB gz (current ~82 KB — Preact compat eval)
- ☐ `vendor-charts` (`recharts`) ≤ 25 KB gz (post-uplot swap)
- ☐ `vendor-auth` (Better Auth + better-fetch + zod) ≤ 30 KB gz
- ☐ `main` ≤ 150 KB gz (current 411 KB — biggest cut)
- ☐ Each lazy route chunk ≤ 60 KB gz
- ☐ `game-xo` ≤ 30 KB gz
- ☐ Total first-paint JS ≤ 250 KB gz on Home (current 490 KB)

If any chunk exceeds budget, the PR fails CI without a `perf-ok` label. Re-bake budgets after Phase 1 lands.

F6 — Optimistic move rendering (perceived-perf, NEW)

The 560–670ms POST → SSE round-trip is the ceiling on Phase 5; below that ceiling the user still feels every move's latency unless the client compensates. The existing Phase 12 ("animation budgets") does not address this.

- ☐ **Render the player's move locally the instant the click lands.** Update the local board state, animate the mark, play sound — all before the POST returns. Reconcile when the SSE `state(moved)` event arrives.
- ☐ **Predict the bot's response.** For built-in minimax bots, the same minimax engine ships in the client bundle. After the player's move, run minimax locally to predict the bot's response, render it as a "ghost" preview after ~150ms, then reconcile with the real SSE event. If predictions match (they will most of the time for minimax), latency feels zero.
- ☐ **Disable click during reconciliation window.** Prevents double-tap from creating an inconsistent local state.
- ☐ **Visual cue if reconciliation rolls back.** A subtle shake + color flash when the local prediction was wrong (rare for built-in bots; meaningful for ML bots).

F7 — Streaming + progressive rendering (perceived-perf, NEW)

The plan mentions "streaming responses for tournament detail" inside Phase 9, but doesn't lay out the priority order or the rendering strategy. Expand:

- ☐ **`TournamentDetailPage` priority cascade.** Render the page header (name, status, time) immediately from URL params or the list-cache. Then `<Suspense>` boundaries in this order: bracket → participant table → match history → coaching cards. Each `<Suspense>` resolves independently so the slowest segment doesn't block faster ones.
- ☐ **NDJSON / chunked transfer for large lists.** `GET /api/tournaments/:id` returns a single JSON body today. Switch to NDJSON: header chunk, then participant chunks, then match chunks. Parse and render incrementally on the client.
- ☐ **Stale-while-revalidate as the default for lists.** When the user navigates back to `/tournaments`, show the cached list *instantly* (we already had it in memory), then refresh in the background. Same pattern for `/leaderboard`, `/bots`.
- ☐ **Skeleton density matches content shape.** Cards with avatar circles, table rows with proportional column widths, bracket shapes that mirror the round-N layout. Misshapen skeletons cause layout flicker on resolve and read as worse than no skeleton at all.

Out of scope (and why)

Option	Reason skipped
SSR / Next.js	Major rewrite. Current SPA already hits FCP ~130ms; SSR's wins (LCP) are addressable cheaper via Phase 3 (image) + Phase 1 (bundle). Revisit only if mobile LCP > 700ms after Phases 1+3.
CDN / Cloudflare in front of Fly	Discussed in v1 — still ruled out for cost / ops; Fly's anycast already gets us most of the way. Revisit when international traffic is non-trivial.
Redis / external cache	Same call as v1 — in-process cache + per-service warmth is enough. SSE pubsub already uses Redis.
Drizzle / Prisma rewrite	Same call as v1 — query rewrite cost > expected gain. Phase 5 (driver) already in.
Move ML training to backend	Tracked separately in `doc/Connect4_Ship_Checklist.md` Spike B — perf is one input but architectural decision is broader.
Tensorflow.js conversion	Same — `doc/Connect4_Ship_Checklist.md` Spike A.

Sequencing *(updated 2026-05-02 from snapshot data)*

The original sequencing was a guess; the snapshot data lets us tier the phases by expected impact. **Tier 0 must land before Tier 1 even gets a benchmark slot** — otherwise the bundle floor masks any other win.

Tier 0 — bundle floor + perceived-move latency (run today)

Two independent dominant costs surfaced from the 0.4 measurements: the JS bundle (cold-page Ready) and the SSE round-trip (every move). Either alone is bigger than anything in Tier 1+.

1. **Phase 1** — Bundle audit + per-route splitting. Concrete sub-steps come from the visualizer; expected to drop `main` from 411 KB gz to ≤ 200 KB gz and Mobile FCP from 1360ms to ~700ms.
2. **Phase 1b** — PlayVsBot deep-dive (independent path bug — different root cause than bundle, but same blast radius for that one user journey).
3. **Phase 5** — SSE round-trip. *Newly promoted from Tier 2 by 0.4 data.* POST move → SSE event arrival is 560ms staging / 670ms prod p50; the gap between POST ack (174ms) and event arrival is ~400ms of pure SSE dispatch latency. Every PvP / PvB move pays this cost. Investigate: SSE flush boundaries, Express response buffering on the SSE channel, Redis pub/sub round-trip if applicable.
4. **Phase 3** — Image diet. *Newly promoted from Tier 1 by 0.4 data.* `colosseum-bg.jpg` is 909 KB on every route. AVIF + responsive sizes + skip-on-mobile is one PR; expected ~700 KB / route savings.
5. **Phase 17** — CI bundle-size guard. Fail PR on > 5% chunk growth. Lock in Tier 0's gains before any new feature work bloats them back.

Tier 1 — confirm budget after Tier 0, then attack what the data still shows

After Tier 0, **re-run `perf-v2.js`** and re-measure. The remaining gap between the new numbers and the budgets table determines what comes next. Likely candidates, not committed:

4. **Phase 0.2** — RUM wiring. Until prod numbers exist, we're guessing from synthetic. Code-only work, ships independently of Tier 0.
5. **Phase 8** — Skeletons everywhere new (TournamentDetailPage, TournamentsPage, BotProfilePage, ProfilePage). Pairs with Phase 1's `<Suspense>` boundaries — the skeleton *is* the fallback.
6. **Phase 3** — Hero image. *Investigate first* (the snapshot showed 0 KB image bytes, suggesting the image loads after Ready). Promote only if DevTools confirms it blocks LCP.
7. **Phase 13** — Tournament page. The heaviest single surface; Phase 1 should already lazy-load it, but the page itself still has a 1900-line render path that needs Suspense splitting + memo / pagination.
8. **Phase 7** — Mobile-specific. Mobile FCP is ~3× desktop today — parse cost on Moto G4 CPU. If Phase 1 doesn't bring it to budget, critical-CSS extraction + transform-only animations are the next levers.

Quick wins — high leverage, can run alongside any tier

These don't need to wait for Tier 0 and don't conflict with later work. Order by gut-feel impact:

- **Service Worker app shell** (Phase 20). Repeat-visit Ready ≈ 0ms. Highest-leverage perceived-perf change in the plan.
- **Speculation Rules API** (Section E1). Native Chromium prerender for same-origin links — zero JS, zero library, instant nav on every cold-anon Chromium visitor.
- **103 Early Hints** (Section E1). Preload critical chunks before TTFB. 50–150ms shaved off cold first paint, free with Fly support.
- **Critical CSS inlined into `<head>`** (Phase 7). Cheap mobile FCP win the platform never had.
- **Materialized view for the leaderboard** (Phase 2). Kills a hot DB path forever; simple Postgres feature.
- **WASM SIMD minimax** (Section E3). Bot move latency from ~30ms to ~1ms. Becomes load-bearing once Connect4 lands.
- **CSS `aspect-ratio` + BlurHash placeholders** (Section E1). Free CLS kill + perceived-instant images.
- **`content-visibility: auto`** on long off-screen sections (Phase 7, Phase 13). One CSS line per container; free render skip.
- **Brotli quality 11** (Section E1) + **Brotli at the edge** (Phase 1). Cumulative 15–25% smaller wire bytes for zero runtime cost.
- **Atomize Zustand stores** (Section E2). One pattern fix; large re-render cut on TournamentDetail under live SSE updates.
- **`scheduler.yield()` in heavy loops** (Section E1). Pulls INP under 100ms on Moto G4 with no algorithmic change.
- **Variable fonts + `font-display: optional`** (Section E1). ~50% smaller font weight + never-blocking render.
- **Pre-warm caches on boot** (Phase 2). Built-in bots, puzzles, system config. The first user stops paying the miss.
- **Resource hints for OAuth providers** (Section E1). Sign-in feels ~200ms faster on cold visitors.

Tier 2 — promote *only* when data shows them as a bottleneck

These were Tier 1 in the original draft but the snapshot doesn't support attacking them yet. Promote to Tier 1 only when re-measurement shows them on the critical path.

- **Phase 2** — DB index pass. Backend p95 wasn't measured yet (Phase 0.3 work). Until RED metrics show DB time > target, this is speculative.
- **Phase 4** — Cross-service network shape. Same — needs measurement before it earns a slot.
- **Phase 5** — SSE channel + POST round-trip. Same — needs measurement, and the user journeys most affected (PvP move latency) aren't on the cold-page benchmark at all.
- **Phase 6** — Backend cold start. Needs cold-start measurement to know whether `auto_stop_machines = false` is worth the cost.

Tier 3 — perceived-perf polish, after the floor moves

These look like wins on a fast platform but reading "Refreshing…" on a 2-second cold page just adds noise. Land after Tier 0 + Tier 1.

- Phase 9 (optimistic / streaming everywhere)
- Phase 10 (hover prefetch on tournament cards / bot rows)
- Phase 11 (route transition polish — view-transitions API)
- Phase 12 (in-game animation budgets)
- Phase 14 (Gym worker-thread inference + WASM)
- Phase 15 (Guide INP audit)
- Phase 16 (Tables-as-primitive overhead)
- Phase 18 (production perf dashboard) — wait for RUM data first.

Tier 4 — architecture experiments (Phase 21 + Section E7)

These are re-architectures, not phases. Pursue only if Tier 0 + Tier 1 + Tier 2 + Tier 3 data still shows the gap they would close.

- **Bun runtime for backend + tournament** (E7). Collapses Phase 6 cold-start entirely; ~30% throughput gain; biggest single win in this tier.
- **Hono on the edge for read paths** (Phase 21 / E7). Promote first if Bun runs into compatibility issues.
- **WebTransport replacing SSE** (E4). Lower latency + better mobile connection migration. Big lift; only after RUM data.

- **WebRTC peer-to-peer for PvP** (E4). Move latency drops to one same-network round-trip once tables are seated. PvP-only payoff.
- **Cloudflare Durable Objects for tablePresence** (E7). Frees backend to scale horizontally.
- **SSR / RSC for `/`** (Phase 21). Cheaper alternatives in Tier 0/1 may close the gap first.
- **Edge functions for hot reads** (Phase 21). Subsumed by Edge KV in E4 if that lands first.
- **gRPC-Web on hot endpoints** (Phase 21 / E7). Revisit after Bun.
- **Self-rolled JWT replacing Better Auth** (E7). Only with measurement showing the auth library is the bottleneck.
- **Preact compat alias** (longshot from Phase 1).

Re-measure cadence

- After Tier 0 lands → run `perf-v2.js --target=staging --warmup` and append a new `Performance_Snapshot_<date>.md`.
- After each Tier 1 phase → same.
- Tier 2 phases require Phase 0.3 RED metrics to be in place first.

Let the data drive every promotion. If a phase lands and the next snapshot doesn't show movement, the phase is *landed but ineffective* and the assumption gets revisited.